20. Construct a C Program to Simulate Reader-Writer Problem Using Semaphores

Code - Part 1 (Lines 1–35) — Reader Function

```
1  #include <stdio.h>
2  #include <pthread.h>
3  #include <semaphore.h>
4  #include <unistd.h>
5
6  sem_t mutex, writeBlock;
7
8  int data = 0;
9  int readersCount = 0;
10
11 void *reader(void *arg) {
12     int i = 0;
13
14     while (i < 10) {
15
16         sem_wait(&mutex);
17         readersCount++;
18
19         if (readersCount == 1) {
20             sem_wait(&writeBlock); // block writer
21         }
22
23         sem_post(&mutex);
24
25         // Reading
26         printf("Reader reads data: %d\n", data);
27
28         sem_wait(&mutex);
29         readersCount--;
30
31         if (readersCount == 0) {
32             sem_post(&writeBlock); // allow writer
33         }
34
35         sem_post(&mutex);
```

Code - Part 2 (Lines 36–67) — Writer Function

```
36
37          sleep(1);
38          i++;
39      }
40
41      return NULL;
42  }
43
44  void *writer(void *arg) {
45      int i = 0;
46
47      while (i < 10) {
48
49          sem_wait(&writeBlock);
50
51          // Writing
52          data++;
53          printf("Writer writes data: %d\n", data);
54
55          sem_post(&writeBlock);
56
57          sleep(2);
58          i++;
59      }
60
61      return NULL;
62  }
63
64  int main() {
65      pthread_t reader_thread, writer_thread;
66
67      sem_init(&mutex, 0, 1);
```

Code - Part 3 (Lines 68–80) — Main Function

```
68      sem_init(&writeBlock, 0, 1);
69
70      pthread_create(&reader_thread, NULL, reader, NULL);
71      pthread_create(&writer_thread, NULL, writer, NULL);
72
73      pthread_join(reader_thread, NULL);
74      pthread_join(writer_thread, NULL);
75
76      sem_destroy(&mutex);
77      sem_destroy(&writeBlock);
78
79      return 0;
80  }
```

Output

```
Reader reads data: 0
Writer writes data: 1
Reader reads data: 1
Writer writes data: 2
Reader reads data: 2
Reader reads data: 2
Writer writes data: 3
Reader reads data: 3
Reader reads data: 3
Writer writes data: 4
Reader reads data: 4
Reader reads data: 4
Writer writes data: 5
Reader reads data: 5
Reader reads data: 5
Writer writes data: 6
Writer writes data: 7
Writer writes data: 8
Writer writes data: 9
Writer writes data: 10


...Program finished with exit code 0
Press ENTER to exit console.
```